

OS Unit 3 Question Bank — Deadlocks

Operating Systems (OS) · Unit 3 · Deadlocks · 2025

Important and frequently asked questions, grouped by marks.

Part A — Short answer (2 marks)

1. What is meant by Deadlock Characterization? Discuss its advantages and limitations.
2. Explain Resource Allocation Graph with a neat diagram and a suitable example.
3. Compare Deadlock Prevention with Banker's Algorithm. When would you prefer one over the other?
4. What is meant by Deadlock Avoidance? Discuss its advantages and limitations.
5. Discuss the role of Banker's Algorithm in Deadlocks.
6. Write short notes on Deadlock Detection and Recovery.

Part B — Long answer (10 marks)

1. Illustrate Deadlock Characterization with a worked example.
2. Explain Resource Allocation Graph with a neat diagram and a suitable example.
3. Explain Deadlock Prevention with a neat diagram and a suitable example.
4. Compare Deadlock Avoidance with Deadlock Characterization. When would you prefer one over the other?
5. Compare Banker's Algorithm with Resource Allocation Graph. When would you prefer one over the other?
6. Explain Deadlock Detection and Recovery with a neat diagram and a suitable example.
7. Discuss the role of Deadlock Characterization in Deadlocks.
8. Illustrate Resource Allocation Graph with a worked example.
9. Compare Deadlock Prevention with Banker's Algorithm. When would you prefer one over the other?
10. Define Deadlock Avoidance. Derive or justify the key properties involved.
11. Write short notes on Banker's Algorithm.
12. Define Deadlock Detection and Recovery. Derive or justify the key properties involved.